

PVL333

Understanding IP

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- As design cost and complexity increase, there is a basic trade-off between the design optimization of the physical product and the cost of the design.
- There are several approaches to the design productivity problem.
- The most basic approaches are purchasing predesigned components and utilizing reconfigurable devices - **Buying IP**
- If the goal is to produce a design optimized in the use of the technology, the fixed costs will be high, so the result must be broadly applicable.
- The alternative to this is to “ reuse ” the existing design.
- These may be suboptimal for all the nuances of a particular process technology, but the savings in design time and effort can be significant.
- *The purchase of such designs from third parties is referred to as the sale of IP .*

Understanding IP

- **The use of IP reduces the risk in design development: It is intended to reduce the design costs and improve the time to market.**
- **The cost of an IP usually depends on the volume.**
- **Specialized SOC designs often use several different types of processors.**
- **Noncritical and specialized processors are purchased as IP and are integrated into the design.**
- **For example, the ARM7TDMA is a popular licensed 32-bit processor or “core” design.**
- **Generally, processor cores can be designed and licensed in a number of ways, as shown in Table.**

TABLE 1.16 Types of Processor Cores Available as IP

Type of Design	Design Level	Description
Customized hard IP	Physical level	IP used in fixed process, optimized
Synthesized firm IP	Gate level	IP used in multiple processes but some optimization possible
Synthesizable soft IP	Register transfer level (RTL)	IP used in any process, nonoptimized

Understanding IP

- **Hard IPs** are physical-level designs that use all features available in a process technology, including circuit design and physical layout.
 - Many analog IPs and mixed-signal IPs (such as SRAM, phase-locked loop) are distributed in this format to ensure optimal timing and other design characteristics.
- **Firm IPs** are gate-level designs that include device sizing but are applicable to many fab facilities with different processor technologies.
- **Soft IPs** are logic-level designs in synthesizable format and are directly applicable to standard cell technologies.
 - This approach allows users to adapt the source code to fit their design over a broad range of situations.
- Clearly, the more optimized designs from the manufacturer are usually less customizable by the user, but they often have better physical, cost–performance trade-offs.